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The World Vegetable Center

SAFER TOMATO PRODUCTION TECHNIQUES

A field guide for soil fertility and
pest management

R. SRINIVASAN
(EDITOR)



AVRDC – The World Vegetable Center



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AVRDC – The World Vegetable Center is an international nonprofit research institute committed to alleviating poverty and malnutrition in the developing world through the increased production and consumption of nutritious, health-promoting vegetables.

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FOREWORD

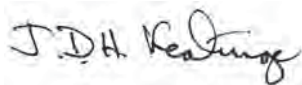
Tomato is one of the most important vegetables in Asia and Africa and these continents account for more than 65% of global tomato production. Tomato is rich in nutrients such as vitamins, minerals, and antioxidants, which are important to well-balanced human diets. Tomato is also an important dietary component because it contains high levels of lycopene, an antioxidant that reduces the risks associated with several cancers and neurodegenerative diseases.

Tomato is susceptible to several insect and mite pests as well as plant diseases. Chemical pesticides are being used indiscriminately to manage these pests in South and Southeast Asia and parts of Africa. In addition, chemical fertilizers and insecticides are sometimes overused in tomato production, which may contaminate groundwater. Intensive agrochemical use in tomato husbandry substantially increases the production cost and may pose serious risks to producers, consumers, and the overall health of the environment.

AVRDC - The World Vegetable Center developed and promoted a safer tomato production strategy in Taiwan from 2005 to 2007. This strategy has reduced reliance on chemical pesticides and optimized the use of organic and inorganic fertilizers. The strategy is being promoted in South Asia, and is ready for broad-scale adoption in other major tomato growing countries in the tropics.

This field guide contains information on healthy seedling production, the optimum use of organic and inorganic fertilizers, and the major insect and mite pests and diseases damaging tomato. It provides integrated pest management strategies for the tropics. Detailed information on major insect pests, mites, and plant

diseases has been provided with relevant pictures. This guide has been prepared for tomato growers and extension specialists. I trust this field guide will be of practical assistance to all tomato growers and extension specialists to produce safer, more wholesome tomatoes in the future.

A handwritten signature in black ink, reading "J.D.H. Keatinge". The signature is written in a cursive, flowing style.

J.D.H. Keatinge
Director General

AVRDC - The World Vegetable Center

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R. Srinivasan

AVRDC - The World Vegetable Center

INTRODUCTION

Tomato (*Solanum lycopersicum* L.) is one of the most widely grown vegetables in the world. It is grown on more than 5 million ha with a production of nearly 129 million t. China is the world's top tomato grower, accounting for more than one-quarter of the world's tomato acreage. Egypt and India together account for more than one-fifth of the world total; Turkey and Nigeria are the other major tomato producing countries. Asia and Africa account for about 79 percent of the global tomato area, with about 65 percent of world output (FAO 2008).

Wild tomato species are native to western South America. There are two competing hypotheses of the origin of domestication of tomato, one supporting a Peruvian origin, the other, a Mexican origin (Peralta and Spooner 2007). Although tomato requires a relatively cool, dry climate for high yield and better quality (Nicola et al. 2009), it is adapted to a wide range of climatic conditions from temperate to hot and humid tropical (Naika et al. 2005). Tomato contains nutrients such as vitamin A, vitamin C, potassium, phosphorus, magnesium, and calcium (USDA 2009); it also contains lycopene, an antioxidant compound that reduces the risk of cancer (Miller et al. 2002).

In the tropics, tomato production is severely constrained by diseases and several insect and mite pests. The major pests include fruit borer, common armyworm, beet army worm, whitefly, leaf miner, and spider mites. The diseases are whitefly-transmitted geminiviruses, bacterial spot, bacterial wilt, damping-off, early blight, late blight, fusarium wilt, southern blight, and black leaf mold. Growers rely heavily on chemical pesticides to protect their tomato crop. For example, farmers in southern

India spray chemical insecticides more than 50 times during a cropping season (Nagaraju et al. 2002). Pesticide misuse has adverse effects on the environment and human health and also increases the cost of production. The share of the cost of pesticide to total material input cost was 31 percent for tomato in the Philippines (Orden et al. 1994). Overuse of chemical fertilizers also causes economic loss to the grower. In addition, excessive use of nitrogenous fertilizers in tomato production has been linked to nitrate contamination of both surface and groundwater (Krusekopf et al. 2002).

This guide provides comprehensive information for tomato growers and extension staff on healthy seedling production, application of balanced and optimum quantities of organic and inorganic fertilizers, and management of major insect and mite pests and diseases. The simple, low-cost integrated pest management (IPM) and soil fertility management techniques outlined in this guide provide satisfactory, sustainable management and can help tomato growers decrease their reliance on agrichemicals. AVRDC - The World Vegetable Center developed, successfully validated, and promoted the safer tomato production strategy in Taiwan from 2005-2007.

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Healthy tomato seedling production

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With the adoption of improved high yielding and disease resistant commercial hybrids, seed has become an expensive input in tomato production. The seedling cost may be significantly increased if growers lose seedlings due to unhygienic practices. This is not uncommon, especially when growers follow the traditional seedbed method to produce seedlings. The following healthy seedling production technique has been validated and proven effective:

- Use locally available seedling trays with hole size about 4 cm deep and 4.5 cm in diameter.
- Fill the holes with a growing medium that drains well, such as peat moss, commercial potting soil, or a mixture of sand, compost and burnt rice hulls. When the latter is used, make sure the compost is mature, and the mixture does not contain any soil-borne pathogens.
- Place the trays on raised areas such as benches in a sheltered place. If benches are not available, prepare raised beds 1.5 m wide and use them as platforms for the seedling trays.
- Seedlings should be grown in a 60-mesh nethouse. If the nethouse is not available, net tunnels should be created over the seedling trays. Erect inverted, “U” shaped, 2 m wide and 1 m high iron or aluminum bars over each bench or raised bed. Maintain a distance of 1 m between two adjacent bars within the row. The thickness of bar should be 1 cm in diameter. Place 60-mesh nylon net over the bar from one end of the row to the other. Pull the net tightly over the bars from all four sides and bury 10-15 cm of net edging in the soil. Make sure there are no gaps between the soil and the net, as these gaps allow insects to get in (Talekar et al. 2003).



- If 60-mesh nylon net is not available, any size up to 32-mesh could be used, but care should be given to monitor the entry of whitefly (e.g., spraying of the outer surfaces of the net with neem or chemical pesticides). If the seedling trays must be kept in open conditions, it is imperative to control insects such as whitefly, thrips, and aphids that transmit diseases. Apply imidacloprid or neem pesticides as seed, soil, or foliar treatments, if necessary.
- Commercially available growing media sometimes may not be properly sterilized. Therefore, use seed treatment with chemical and/or biocontrol agents as a prophylactic measure for managing soil-borne diseases. For instance, the seed can be treated with broad-spectrum fungicides, such as captan and/or thiram to reduce losses from damping-off (Hanson et al. 2000). Alternatively, seed may also be treated with *Trichoderma viride* and *Pseudomonas fluorescens*.
- Sow two seeds per hole at a depth of 0.5 cm and thin the seedlings 2-3 days after the first true (non-cotyledon) leaves appear (Hanson et al. 2000).
- Water at the rate of 15 ml per hole during first irrigation. Subsequent irrigation should be given at the rate of 7.5-10 ml (maximum) per hole daily (morning hours preferred). If the temperature is higher, especially during summer, apply 7.5-10 ml (maximum) per hole twice a day—once in the morning and once in the afternoon. The amount of water required depends on the growing substrate and its moisture-holding abilities.
- Seedlings will emerge within eight days at optimal soil temperatures of 20 - 30°C (Hanson et al. 2000).
- If the seeds were not treated with chemical or biological agents, apply a locally available and recommended fungicide (e.g. Etridiazole) to control soil-borne fungal diseases. Follow the





recommended dose, and apply at the rate of 5 ml solution (fungicide and water) per seedling. Apply on the growing substrate one to two days prior to transplanting.

- After three weeks, check the seedlings for vigor and leaf color. If the leaves turn slightly yellow or seedlings seem thin, apply fertilizer with NPK ratio of 15-10-15 + 2 MgO through foliar or soil application. Dilute the fertilizer 1000 times and apply at the rate of 5 ml per seedling. Apply 1-2 times before transplanting. Monitor seedling growth; if the seedlings are growing too rapidly before you are ready to transplant, apply less fertilizer.
- Remove the net and apply the fungicides or fertilizers as quickly as possible. Close the net again promptly to prevent insects from entering. If the seedling trays are kept in several benches or beds, open only one bench or bed at a time.
- Use good seedlings at the four- or five-leaf stage (about 4 weeks old), vigorous and stocky for transplanting (Hanson et al. 2000).

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Soil fertility management for safer tomato production

Ma Chin-Hua and Yueh-Huei Lin

Crop and Ecosystem Management

AVRDC - The World Vegetable Center





Overuse and imbalanced use of manure and inorganic fertilizers in vegetable production systems have reduced yields of vegetables and caused environmental pollution as well as human health hazards. Inadequate fertilizer use, improper fertilizer management, and depletion of soil nutrients result in low yields and cause land degradation in many developing countries. Balanced and efficient use of plant nutrients increases productivity and profits and minimizes environmental risks.

Over-application of fertilizer

Excessive application of both organic and inorganic fertilizers in intensive vegetable production systems is very common in many countries. However, not all nutrients applied to the soil are taken up by plants. Unused nutrients may remain in the soil and become an environmental hazard through water leaching in soil or in runoff, or be lost to the atmosphere by volatilization.

All water soluble nitrogen (N) fertilizers are fairly rapidly oxidized to nitrate after application into soils under normal farming conditions. Nitrate is not adsorbed by soil particles, and is therefore the most likely element to be leached out into surface water or groundwater, or to be lost to the atmosphere by denitrification. Nitrate derived from the decomposition of organic matter in soil or manure is also subject to leaching. Leaching of NO_3^- to groundwater may increase its content in drinking water, and lead to higher accumulations in plant tissues. The consumption of excessive NO_3^- in drinking water and food may cause methemoglobinemia; infants under 6 months of age are particularly susceptible to this health hazard (“blue-baby syndrome”). The potential carcinogenic effect of nitrosamines from N is a major human health concern. Ammonia volatilization may occur from manure



heaps and from soil soon after manure application. Ammonia also can be volatilized if urea fertilizer is applied to high pH soils under hot and dry conditions.

Phosphorous (P) is supplied to plants in the form of phosphate ions. They are relatively immobile in the soil, being strongly adsorbed by clay particles, and react with the surfaces of iron, aluminum, and manganese oxides and hydroxides. Applied phosphate not taken up by the crop remains in the soil and may be washed off the field by runoff or soil erosion. Excess amounts of phosphates and nitrogen in surface water lead to eutrophication, which causes algal blooms in surface waters. When algae and water plants die, the bacteria that decompose them increase rapidly and consume the dissolved oxygen in the water. Many fish and aquatic species cannot survive when the dissolved oxygen content decreases. Algal blooms pose serious problems for ecosystems and human society.

Potassium (K) ions with positive charge can be adsorbed by the soil and can even be fixed within clay particles. However, potassium that is not taken up by the plants may be leached out or run off. Major losses of potassium are caused when the element is washed away in liquid manure from farmyards and dairies. Potassium in water has no detrimental effects on health, but may indicate the presence of animal effluent or overuse of fertilizers. It is not a major factor in the eutrophication of water.

Imbalanced fertilizer application

Nitrogen usually shows significant effects soon after application: plants grow more vigorously with a dark green leaf color. However, too much nitrogen in vegetables may result in plant lodging, greater weed competition, and more pest problems, with loss of yield and decrease in





storage quality. The nitrogen not taken up by the crop is likely to be lost to the environment.

Farmers expecting an immediate and evident effect supply crops exclusively with nitrogen (e.g., urea), which is available at an affordable price. However, higher yields of vegetables will also take up greater amounts of phosphorus and potassium from the soil. Thus, increased yields through the application of nitrogen alone deplete the other plant nutrients in the soil. Research has shown that nitrogen efficiency declines without phosphorus and potassium application. Balanced fertilizer application with adequate supplies of nitrogen, phosphorus, and potassium to meet plant requirements and expected yield of the crop is crucial for safe vegetable production.

Inadequate nutrient inputs

The continuous cropping and constant removal of soil nutrients without sufficient replenishment causes a steady decline in soil fertility and leads to soil degradation and desertification. Organic recycling is not sufficient because the biomass produced in depleted soils is extremely poor in essential nutrients. Minimum amendments and adequate nutrient inputs are essential to make these soils productive and sustainable.

Fertility management for safer vegetable production

1. Add nutrients to replenish stocks in the soil

Use a combination of organic and inorganic fertilizers to improve soil fertility: The inorganic fertilizer provides higher available nutrients to increase productivity in the short term, while the organic fertilizer increases soil



organic matter, soil structure, and soil buffering capacity to sustain soil fertility in the long term. Balanced fertilizer application using a combination of organic and inorganic sources may improve the sustainability and profitability of crop production. Long-term application of inorganic fertilizer may cause soil acidification, whereas long-term application of certain organic fertilizers may increase soil pH to alkaline conditions. A balanced application of organic and inorganic sources is important to sustain good soil conditions and make the nutrient supply more efficient. A ratio of 1:1 or 1:3 organic to inorganic fertilizer based on N requirement is a common recommendation to farmers.

Blend fertilizers to meet the uptake ratio of major nutrients in crops: Balanced fertilizer application also should be considered between ratios of N, P, and K. Many organic fertilizers are low or high in N, P, or K. Application of organic fertilizer based on N requirement may result in K or P accumulation. However, inorganic compound fertilizers with equal composition of NPK, such as 15-15-15 or 20-20-20, are sometimes the only source of inorganic fertilizer in many countries. For most crops, the P uptake is only 1/5~1/10 of that of N or K uptake. For example, as shown in Fig. 1, total nutrient uptake of NPK by tomato is 9.5-12-13.5 g/plant at 120 days after transplanting. By conversion, the uptake ratio of tomato is equal to $N:P:K = 1:0.13:1.42$. This data shows that tomato plant has a greater potassium requirement compared with other vegetables. Continuous application of organic and inorganic fertilizer with imbalanced or equal ratio of NPK may result in certain nutrient accumulations in soils. It is recommended that farmers improve fertilizer application by mixing several kinds of organic fertilizers to form a desired ratio that will best fit the crop cultivated. Farmers should also apply different





combinations of inorganic fertilizers during the growth period to meet plant requirements and reduce nutrient accumulation and leaching in soils.

2. Improving nutrient use efficiency

Add fertilizers synchronized with the growth patterns of tomato plants: Every crop has a unique dry matter accumulation pattern as well as nutrient uptake patterns. Fertilizer application timing and amounts matched with the growth pattern of crops are crucial for enhancing fertilizer efficiency. Fig.1 shows the nutrient uptake pattern of tomato. N requirements are moderate during vegetative growth until fruit set. P is very important for vigorous growth and fruit production. K is required for fruit set and enlargement. Important growth stages for nutrient uptake include seedling establishment, when maximum nutrient uptake occurs, and early fruiting, when the maximum rate of K accumulation takes place. Both organic and inorganic

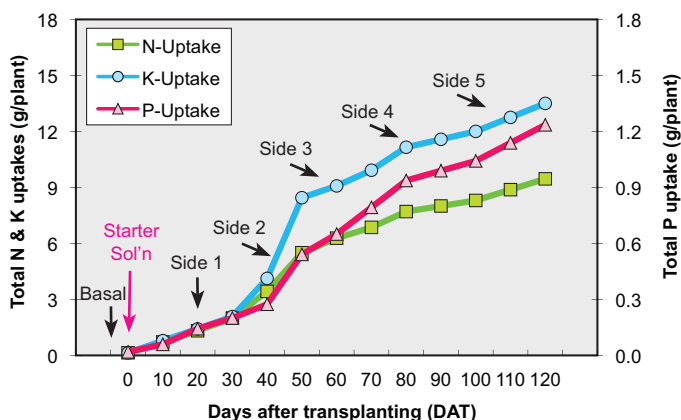


Fig.1. N, P, and K uptakes in the tomato plant over a 120-day growing period. (yield potential 60 t/ha)

fertilizers are applied as basal before transplanting, and inorganic fertilizers are applied as side-dressing at 20, 40, 60, 85, 105 days after transplanting. During fruit set and the enlarging stage, side-dressing should be applied more frequently. N leaching depends on the fertilizers applied, timing of precipitation, irrigation, soil type and plant N uptake behavior. Split application of N fertilizers during a single cropping period is recommended for controlling N, especially NO_3^- leaching into groundwater.

Enhance efficiency through new fertilizer application practices: Most vegetables require high amounts of nutrients in a relatively short growth period. Sustaining adequate NPK concentrations in the soil solution during active growth periods is crucial for increasing crop productivity. AVRDC - The World Vegetable Center has developed “Starter Solution Technology” for several vegetables: Small amounts of concentrated inorganic fertilizer prepared as a liquid solution are applied immediately after transplanting and as well-timed side-dressings for tomato. Starter Solution provides vital nutrients to young tomato plants before their root systems are well-established and helps plants to meet their immediate nutrient needs during active growth periods, resulting in higher fruit yields. Starter Solution can be prepared by diluting inorganic liquid compound fertilizer (14%N-28% P_2O_5 -14% K_2O) and applying the solution at a rate of 2.4 g (240N-210P-200K mg) in 50 ml water per plant (equivalent to 7.2N-6.2P-6K kg/ha) after transplanting, for cherry tomato in a sandy loam soil. The solution should be applied in a volume less than 1% of soil maximum water holding capacity and usually will be adsorbed by the soil surface near plant roots. Although the concentration is very high, the concentration decreases near the root vicinity after reaction with soil, and the solution maintains a high level of nutrients in the soil.





The application of Starter Solution significantly boosted early growth and overall yield of all vegetables that we tested. It also enhanced the release of nutrients from organic composts. The booster effects of Starter Solution application on initial plant growth are evident: An application of 7.2N-6.2P-6K kg/ha starter solution could substitute 30~50% of inorganic fertilizer and half the amount of organic fertilizers. It also reduced residual mineral N in soil, which might easily cause environmental pollution after cultivation. Our research showed that maximum yields of cabbage, cherry tomato, and chili pepper were obtained using a basal application of manure compost, an application of Starter Solution at transplanting, and then followed by various side-dressings of supplemental fertilizers depending on crop and season. Balanced fertilizer application practices based on Starter Solution technology in combination with organic and inorganic nutrient sources were found to increase fertilizer efficiency, increase farmer profits, and reduce environmental pollution.

3. Reduce nutrient losses in soil

A reduction in fertilizer amounts is an effective way to decrease nutrient leaching. Proper arrangement of the cultivation schedule is another way to avoid nutrient loss. For example, green manure decomposes very fast after incorporation into soils under tropical conditions. Therefore, it is necessary to have a successive crop in the field after a few weeks of incorporation to use nutrients released from the green manure. Mulching beds and covering crops in the field during rainy seasons are also effective practices to avoid run-off and nutrient losses.



4. Better management of available resources in the system

Add organic matter to soil: Organic matter turnover is very rapid in tropical soils, and maintaining an adequate level of organic matter in the soil is a difficult challenge. There are many ways to build up soil organic matter, such as fallowing, growing a green manure crop, adding farmyard manure, compost or organic fertilizers, incorporating crop residues, etc. All of these are promising methods to improve soil organic matter contents. In recent years, increasing concern about the environment and soil sustainability has prompted interest in composting as a way to recycle wastes back to soil. Extensive research has developed science-based and environment-friendly composting technologies. Composting can be enhanced by adjusting C/N ratio to 20-30:1, maintaining moisture at 50-60%, and providing good aeration to encourage the growth of microorganisms. Proper composting increases the nutrient efficiency in the compost, and also decreases N losses during the composting process.

Adopt proper crop rotation: Proper crop rotation sustains soil organic matter, increases soil cation exchange capacity (CEC), and improves soil biological properties that may be able to suppress some soil-borne pathogens. It is recommended that green manure crops be included in the system, especially species with N fixing capacity. Solanaceous vegetables should not be cultivated on the same piece of land without rotation. Soil fallow is important to recover soil fertility after intensive cultivation. Crop rotation of rice with upland vegetables should be managed carefully, because rice cultivation may destroy soil structure and does not support succeeding vegetable cultivation.





Fertilizer recommendation practices for safer tomato production

Calculate nutrient uptake/removal by tomato plants based on target yield: Fertilizer recommendations depend on local conditions. Table 1 lists data of nutrient uptake/removal by tomato fruit, indicating variations that exist among different data sources. Optimum fertilizer recommendations for each region should be determined in cooperation with local experimental stations, fertility management specialists, and leading farmers, or from your own fertilizer trials, to determine optimum rates.

Table 1. Nutrient uptake/removal of tomato fruit estimated based on target yield

Target yield	Nutrient uptake/removal (kg/ha)			Data Sources*
(t/ha)	N	P ₂ O ₅	K ₂ O	
24	177	46	319	IFA, tropic conditions, various sources
40-50	100-150	20-40	150-300	IFA, temperate conditions
40	108	31	164	AVRDC, processing tomato, 2005
35	87	30	127	AVRDC, cherry tomato, 2001
40	132	37	202	IPNI, nutrient removal in selected vegetables, 1996-2007
50	140	65	190	IPNI, Dr. H. L. S. Tandon. Fertilizer Recommendation for Horticultural Crops, 2000
40	110	30	150	FAO, Fertilizer and their use, 2000

* IFA: International Fertilizer Industry Association
 IPNI: International Plant Nutrition Institute
 AVRDC: AVRDC - The World Vegetable Center



The potential target yields of tomato for a particular location should be identified from previous years' data. The amount of fertilizer to be applied depends on soil fertility, fertilizer recovery rate, soil organic matter, mineralization, and leaching of N. A soil test is recommended to determine the available N, P, and K. The amount to be applied can then be calculated based on the target yield and available residual nutrients. When harvesting tomato fruit, the nutrients taken up by the fruit are removed from field. Safe and rational fertilizer estimation also should be based on nutrient removal by the fruit. Consider the data from IPNI as an example: nutrient uptake of NPK is equivalent to 3.3-0.4-4.2 kg/t of tomato fruit harvested. If we assume a potential tomato yield of 40 t/ha, then total nutrient removal to achieve the target yield is estimated by multiplying individual nutrients by 40, i.e., NPK = 132-16-168 kg/ha (Table 2, equivalent to N-P₂O₅-K₂O = 132-37-202 kg/ha, Table 1).

Convert to nutrient requirement by fertilizer recovery factor: The amount of fertilizer required should be converted from nutrient removal by fertilizer recovery factor. However, fertilizer uptake efficiency by a crop is highly variable and depends upon many factors, including fertilizer form, timing, and placement, as well as irrigation, precipitation, soil type, and other management practices. In the tropics, fertilizer recovery rate is generally low. If we assume the fertilizer recovery rate of NPK = 40%-10%-50%, total nutrient requirement for a target yield of 40 t/ha (fresh weight) is listed in Table 2.

Adjust actual fertilizer requirement by soil test: The actual fertilizer requirement needs to be adjusted downward because some of the nutrients can be supplied from the soil. The amount of NPK already in the soil can be





estimated by soil testing. Addition of fertilizer is needed to make up the difference between the NPK fertilizer requirement for the target yield and the NPK available in the soil. For example, if the soil test indicates that 100 kg each of N, P, and K are available, you would need to apply only 230N-60P-236K kg/ha, which is equivalent to $N:P_2O_5:K_2O = 230-138-283$ kg/ha. This application amount includes nutrients from both organic and inorganic fertilizers.

Table 2. N, P, and K requirements, fertilizer recovery rate, and amounts to apply for a target yield 40 t/ha of tomato.

	Nutrient element		
	N	P	K
Nutrient removal (kg/t yield)	3.3	0.4	4.2
Nutrient removal (kg/ha)	132	16	168
Fertilizer recovery rate (%)	40	10	50
Amount needed (kg/ha)	330	160	336
Nutrient available in soil* (kg/ha)	100	100	100
Actual fertilizer applied (kg/ha)	230	60	236

* Assumes nutrients available in the soil; the actual fertilizer amount applied should be adjusted downward based on the soil testing results.

The values from a soil test do not show fertilizer requirements directly. Classifying soil test values as “high,” “medium,” and “low” does not indicate how much fertilizer needs to be applied to get the desired and most economical yield increase. To make a good fertilizer recommendation, results of soil tests must be calibrated with crop response. When a soil is considered “very high” in fertility, actual fertilizer requirement can be adjusted downward to 30~40% of the calculated amounts, or to 60~70% when a soil is classified as “high” in fertility.



Partition the fertilizer - different sources and times:

Tomato plants should be fertilized with organic and/or chemical fertilizers to produce high yields. Compost and/or manures help increasing soil organic matter content, which increases the soil's buffering capacity and nutrient retention ability. The fertilizer amounts mentioned above include organic and inorganic fertilizers. Thirty percent of the actual fertilizer amount should be applied as organic compost or manure (8-10 t/ha), while the remaining amount should be applied as inorganic fertilizer ($\text{N:P}_2\text{O}_5:\text{K}_2\text{O} = 161\text{-}96\text{-}199$ kg/ha).

N

Thirty percent of the inorganic N should be applied as basal fertilizer before transplanting. Four percent of N should be applied as Starter Solution immediately after transplanting. The remaining N should be side-dressed in three equal amounts at 3, 6, and 9 weeks after transplanting (WAT).

P

Thirty percent of the P should be applied as basal fertilizer before transplanting. Fifteen percent of P should be applied as Starter Solution immediately after transplanting and the remaining P should be side-dressed at second side-dressing (6 WAT).

K

Twenty percent of K should be applied as basal and 3 percent should be applied as Starter Solution. The remaining K should be divided into three splits, applied at the same time with N side-dressings (Table 3).

The rates and timing of fertilizer application need to be modified to take into account lower yield, longer growth duration, and different nutrient uptake per unit yield. If fertilizer availability is a constraint, farmers can apply





less fertilizer, but should maintain the NPK ratio and partitions as suggested above.

Table 3. Fertilizer partitions, application timings, and application method for tomato production

Application Timing	Application Method	Fertilizer Sources	Fertilizer rate (kg/ha)		
			N	P ₂ O ₅	K ₂ O
Basal (pre-planting)	Band	Manures or composts ^c , 10 t/ha	70	45	85
Basal (pre-planting)	Band	Inorganic fertilizer	50	30	40
Starter Solution (concentrated) ^a	Soil near base of plant	Inorganic fertilizer	6	14	6
1st side dress (3 WAT) ^b	Spot	Inorganic fertilizer	35	--	50
2nd side dress (6 WAT)	Spot	Inorganic fertilizer	35	50	50
3rd side dress (9 WAT)	Spot	Inorganic fertilizer	35	--	50

a Starter Solution can be prepared with locally available soluble fertilizers with all NPK at a rate of 180-240 mg N/50 ml/plant. At first time of application, test the concentration to find out the optimum rate for your tomato plants.

b WAT = weeks after transplanting, timing for the side-dressing can be changed based on growth of the plants. If fruit yield is high, additional side-dressing is recommended after 1st harvest of fruit.

c Assumes the manure contains 1.27-0.82-1.55% of N-P₂O₅-K₂O and with 55% dry matter content.



Supply other important plant nutrients: Tomatoes are sensitive to excess or deficiency of macro- and micronutrients. K deficiency is likely to occur in many locations due to insufficient or imbalanced K in many countries. Fruit of K-deficient plants ripen unevenly, appear yellowish and blotchy, and are thin-fleshed. Calcium (Ca) deficiency causes blossom-end rot of tomato fruit, usually induced by water stress in the plant. At time of fruit set, cells at the blossom end cannot receive sufficient Ca due to poor Ca translocation to the flowers, resulting in dry-rot brown spots on the bottom of enlarging fruit. Ca deficiency is most likely to occur on strongly Ca-depleted tropical soils, sandy soils, or acid soils. Ca can be supplemented through foliar fertilizer application or liming.

When magnesium (Mg) deficiency occurs, older and lower leaves are first affected; leaf veins remain dark green, while areas between the veins become yellow. Mg deficiency is likely to occur on sandy soils with low cation exchange capacity, low Mg-supplying soils, acid soils, and in the presence of high levels of K. Apply Mg-containing fertilizers or dolomite limestone to Mg-deficient soils.

Most sulfur (S) in the soil is tied up in organic matter and cannot be used by the plant until it is converted to the sulfate (SO_4^{2-}) form by soil bacteria. Sulfur is not mobile in plants; when deficiency occurs, new growth suffers first. Plants show a pale green coloring of younger leaves. Sulfur can be broadcast, or banded in S-containing fertilizers or sulfur powder.

Deficiencies of iron (Fe), zinc (Zn), manganese (Mn) and boron (B) are likely to occur in calcareous soils. Tomato should be supplied with these nutrients when the soils are deficient of these particular nutrients. Iron (Fe) is taken up by the plant as Fe^{+2} cation. When Fe deficiency occurs,



Fertilizer application tips for safer vegetable production

- Select a tomato variety well-adapted to your local conditions and with high nutrient use efficiency.
- Blanket application of fertilizer leads to over-application. Instead, follow a balanced fertilizer application scheme based on estimates of nutrient removal by fruit.
- Nutrients removed from the soil through harvesting of produce should be replenished from both organic and inorganic sources to maintain productivity.
- Split side-dressing into three or four applications, rather than just one. This practice increases nutrient use efficiency and reduces losses.
- Maintain balanced NPK fertilizer application, especially for tomato, which requires more K than N.
- Use Starter Solution for safer tomato production.
- Make the best use of whatever quantity of fertilizer you can afford to maximize returns and maintain the safety and sustainability of your production system.

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Insect and mite pests on tomato: identification and management

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Several kinds of insect and mite pests attack tomato leaves, flower buds, and fruits during tomato cultivation. The most commonly occurring pests on tomato are fruit worm or fruit borer (*Helicoverpa armigera* Hübner), common army-worm (*Spodoptera litura* Fabricius), beet army-worm (*Spodoptera exigua* Hübner), whitefly (*Bemisia tabaci* Gennadius), leaf miner (*Liriomyza* spp.) and two-spotted spider mite (*Tetranychus urticae* Koch). However, whitefly and fruit worm are the major insect pests in most of the tropical tomato-producing regions of the world, as whitefly transmits leaf-curl virus disease and fruit worm causes severe damage to the fruit, reducing the marketable yield of tomato.

Tomato fruit worm

Helicoverpa armigera Hübner
(*Lepidoptera: Noctuidae*)

The tomato fruit worm, a polyphagous and highly mobile insect, is a pest of economic importance on many agricultural and horticultural crops. It has attained the status of major pest on a number of crops, including cotton, tobacco, corn, sorghum, sunflower, soybean, Lucerne, and pepper (Torres-Villa et al. 1996). It has been recorded as a damaging pest on 180 cultivated and wild plant species in at least 45 families (Venette et al. 2003a).

Biology

The adult is a stout-bodied moth with a wingspan of about 35-40 mm (Plate 1). The adult males are usually pale yellow with olive green or gray color; the adult



females are reddish-brown. The male forewings are pale yellow and olive green in color with light brown obscure transverse lines; female forewings are reddish-brown with distinct brown-black transverse lines. The hind wings are white with a brown border. Mostly, the female moths emerge first, and release sex pheromones to attract males 2-5 days after emergence. Mating occurs 1-4 days after emergence. Female moths generally live longer than males. In the laboratory, longevity varied from 1-23 days for males and 5-28 days for females (Pearson 1958). Bhatt and Patel (2001) recorded a slightly longer lifespan of about 51 days for males and 54 days for females.

The adults feed on nectar and lay eggs singly and scattered, usually on or near the leaflets, floral buds, or young fruit. They prefer to lay eggs on the hairy surfaces of the plant. Maximum egg-laying coincides prior to or during host flower production (King 1994). On average, a single female moth can lay about 730 to 1702 eggs, with a maximum of 4394 eggs in an oviposition period of 10 - 23 days (King 1994; Fowler and Lakin 2001; CAB 2003). The eggs are spherical shaped, about 0.5 mm in diameter, creamy white when laid, but later turning brown, then black just before hatching. The egg period is about 4-5 days, depending on the temperature. Eggs hatch in about 3 days at 25°C, but at lower temperatures may extend up to 11 days (CAB 2003).

Upon hatching, the neonate larvae are creamy white with dark brown or black colored heads and prominent spines on the body. The grown-up larvae vary in color, ranging from pale green to brown or even black with lateral stripes on the body (Plate 2). Mature larvae can grow to a size of about 40 mm in length. Larval period is about 15-25 days, depending on the temperature, host plant, and several other factors. The larva passes through five to





seven instars, with six being most common. Later larval stages are found singly on fruit or other plant parts. If crowding occurs due to high larval density, cannibalism can be observed among the grown-up larvae. The pre-pupal stage lasts for 1-4 days, and during this time larval activity decreases (King 1994).

Pupation takes place in soil, at a depth of 2.5-17.5 cm. Sometimes pupation takes place on plant or soil surfaces (King 1994). Pupae are dark brown (Plate 3). The pupal period varies from 6-33 days depending on the temperature; on average, it is about 10 days to two weeks. The optimum survival temperature for pupa was 27°C (Twine 1978). Little or no diapause is observed in tropical areas (King 1994). However, the pupae may enter into diapause depending on the photoperiod and temperature. Diapause occurs when larvae are exposed to day lengths of about 11.5-12.5 hours, and low temperatures (19-23°C), or when larvae are exposed to lengthy periods of extremely hot and dry weather ($\geq 35^{\circ}\text{C}$) (King 1994; Zhou et al. 2000; Shimizu and Fujisaki 2002; CAB 2003). In a laboratory study, high temperatures (above 37°C) caused pupal dormancy (Nibouche 1998).

Damage symptoms

The neonate larvae feed on the surfaces of leaves or floral buds. However, the grown-up larvae prefer to feed on the contents of reproductive parts such as floral buds, flowers, and young fruit. The larvae make holes in these parts and feed by thrusting their heads inside (Plate 4). The holes are circular (Plate 5) and often surrounded by fecal pellets. Later, the larva feeds on most of the inner contents of the fruit, hollowing it out. Severely damaged fruit rots and falls down; partially damaged fruit may become deformed.



Plate 1: *Helicoverpa armigera* - adult

Plate 2: *Helicoverpa armigera* - larva

Plate 3: *Helicoverpa armigera* - pupa

Plate 4: *Helicoverpa armigera* damage on young fruit

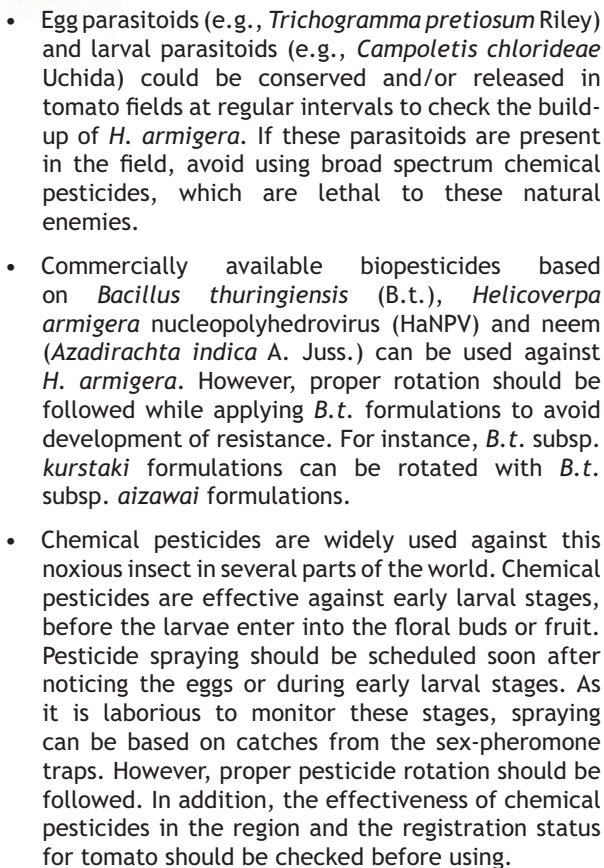
Plate 5: *Helicoverpa armigera* damage on mature fruit



Management

- **Monitoring:** Sex pheromone traps baited with *H. armigera* pheromone lures trap the adult male moths. Trapping can be used to predict population build-up in the field.
- **Mass-trapping:** Sex pheromone traps baited with *H. armigera* pheromone lures can be used to trap as many males as possible to reduce chances of females mating and producing viable eggs in the field. However, this has limited effect for polyphagous insects like *H. armigera*; its population is always higher due to the availability of multiple host plants in crop production systems in the tropics.
- **Mating disruption:** High concentration of full blends or, at times, only one component of the multicomponent pheromone is placed in the field to permeate the area. The high concentration of pheromone in the air overwhelms the male, making it impossible to locate a receptive female. Failure to mate results in failure to produce eggs, or the production of only nonviable eggs, thus reducing the build-up of insect populations. Placing a high concentration of sex pheromone in a slow-release formulation on a 5- and 10-m grid in the field results in a drastic reduction in male moths attracted to virgin females, which adversely affects mating in *H. armigera* (AVRDC 1988).
- Planting African marigold (*Tagetes erecta* L.) as a trap crop on both sides and parallel to 10 and 15 rows of tomato, respectively reduces the incidence of *H. armigera* (Srinivasan et al. 1994). *H. armigera* adults preferred marigold at flowering stage over tomato for oviposition. This reduced *H. armigera* infestation of tomato. It is important to synchronize transplanting of both crops so that flowering coincides, which attracts *H. armigera* female adults. As tomato can continuously produce flowers over a long growth period, the marigold must also have flowers during the period.





Common armyworm

Spodoptera litura Fabricius

(*Lepidoptera: Noctuidae*)

Like *H. armigera*, *S. litura* is also a polyphagous and highly mobile insect and a pest of economic importance on many agricultural and horticultural crops. The larvae are nocturnal and feed actively during night hours. During the day, the larvae hide under the soil, in cracks and crevices, in plant debris in the field.

Biology

Adults of *S. litura* are very similar in size and coloration to *S. ornithogalli* (found in North and Central America) and *S. littoralis* (found in the Mediterranean, Middle East, and Africa) (Mochida 1973; IIE 1993; Venette et al. 2003b). However, *S. litura* is the predominant species on tomato in tropical South Asia and Southeast Asia. There is no overlapping in the geographical distribution between *litura* and *ornithogalli* or *littoralis* in this region. *S. litura* adult is a stout-bodied moth (Plate 6) with a wing span of about 40 mm. The adults are usually brown in color; the forewings have numerous criss-cross streaks on a cream or brown background. The hind wings are white with a brown patch along the border. The eighth abdominal segment of female moths possesses dense scales. The eggs are laid in groups of 200-300, and covered with brown hairs from the body of the mother (Plate 7). The egg period is about 3-5 days.

Upon hatching, the neonate larvae are translucent green with dark thorax and gregarious (Plate 8). The young larvae remain and feed in groups. However, they disperse when older to feed individually. The grown-up larvae are green, pale greenish brown, or black in color, with stout,





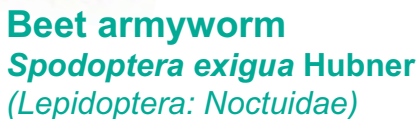
Plate 6: *Spodoptera litura* - adult

Plate 7: *Spodoptera litura* - egg mass

Plate 8: *Spodoptera litura* - early larvae in group

Plate 9: *Spodoptera litura* - grown-up larva





Like *S. litura*, *S. exigua* is a polyphagous and nocturnal insect that feeds on tomato, eggplant, hot and sweet peppers, onion, etc.

Biology

Adults of *S. exigua* are medium-sized moths with a wing span of about 30 mm. The adults are usually brown in color; the forewings are mottled brown. The hind wings are gray with a brown line along the border. The adult period is about 10 days, and on average a single female lays about 500-600 eggs.

The eggs are laid in groups of 100-150 and covered with the brown hairs from the body of the mother similar to *S. litura*. The egg period is about 3-5 days.

The grown-up larvae are brownish green dorsally and pale yellow colored ventrally with a white or yellow lateral stripe (Plate 10). Larval period is about two to three weeks. The larva passes through five instars.

Pupation takes place in earthen cocoons in soil. Pupae are light brown. The pupal period varies from 7-11 days.

Damage symptoms

The larvae cause defoliation. Sometimes, the larvae may cut the seedlings or young plants at soil level. Occasionally the larvae may feed on the fruit, boring single or multiple holes in the flesh.

Management

Similar to *S. litura*.





Plate 10: *Spodoptera exigua* - Larva





Whitefly

***Bemisia tabaci* Gennadius**

(Hemiptera: Aleyrodidae)

The whitefly is widely distributed in tropical and subtropical regions, and in greenhouses in temperate regions. *B. tabaci* is highly polyphagous and is known to feed on several vegetables including tomato, eggplant and okra, field crops and weeds. Hot, dry conditions favor the whitefly, and heavy rain showers drastically reduce population build-up. This insect is active during the day and settles on lower leaf surfaces at night.

Biology

The whitefly adult is a soft-bodied, moth-like fly (Plate 11). The wings are covered with powdery wax and the body is light yellow in color. The wings are held over the body like a tent. The adult males are slightly smaller in size than the females. Adults live one to three weeks. The females mostly lay eggs near the veins on the lower surface of tomato leaves. They prefer hairy leaf surfaces to lay more eggs. Each female can produce as many as 300 eggs in its lifetime.

Eggs are small (about 0.25 mm), pear-shaped, and vertically attached to the leaf surfaces through a pedicel. Newly laid eggs are white and later turn to brown (Plate 12). Egg period is about 3-5 days during summer and 5-33 days in winter (David 2001).

Upon hatching, the first instar larva (nymph) moves on the leaf surface to locate a suitable feeding site. Hence, it is commonly known as a “crawler.” After identifying a suitable feeding site, it will insert its piercing and sucking mouthpart and start sucking the plant sap from the phloem. This first instar nymph has antennae, eyes

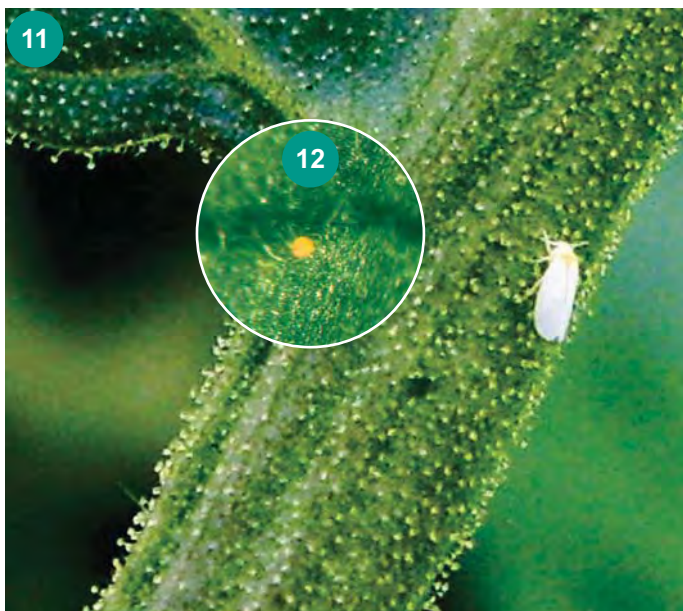


Plate 11: *Bemisia tabaci* - adult

Plate 12: *Bemisia tabaci* - egg





and three pairs of well-developed legs. The nymphs are flattened, oval-shaped and greenish yellow in color. The legs and antennae are atrophied during the next three instars and they are immobile during the remaining nymphal stages. The last nymphal stage has red eyes (Plate 13), and is sometimes known as puparium, although insects of this order (Hemiptera) do not have a perfect pupal stage (incomplete metamorphosis). Pupal period is about 9-14 days during summer and 17-73 days in winter (David 2001). Adults emerge from puparia through a T-shaped slit and the empty pupal case is known as exuvia.

Damage symptoms

Direct damage: Both the adults and nymphs suck the plant sap and reduce the vigor of the plant. In severe infestations, the leaves turn yellow and drop off. When populations are exceedingly high (Plate 14) they secrete large quantities of honeydew, which favors the growth of sooty mold on leaf surfaces and reduces the photosynthetic efficiency of the plants.

Indirect damage: *B. tabaci* acts as vector for several viral diseases including tomato yellow leaf-curl virus (TYLCV). Plants infected by TYLCV show stunted growth with erect shoots. Leaflets curl upwards and inwards, are reduced in size, stiff, thicker than normal, and leathery in texture (Plate 15). Leaves have yellowing along the margins and interveinal chlorosis (yellowing). The flowers wither and droop, and fruit set is reduced or nil. Fruit, if produced, is small and unmarketable. The nymphs acquire the virus while feeding on the plant sap from a TYLCV-infected plant; the virus persists in the insect, which can then transmit it throughout its life. It has been proven that the virus can be transferred from mother flies to the offspring.



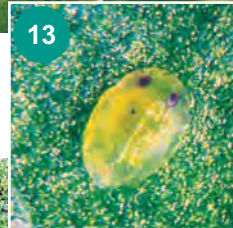
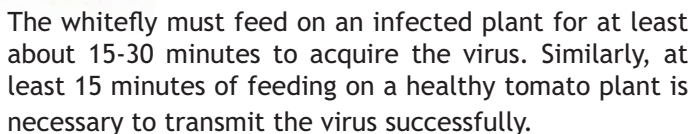


Plate 13: *Bemisia tabaci* - red-eyed nymph
 Plate 14: Colony of *Bemisia tabaci*



Plate 15: Tomato plant infected by leaf-curl virus (TLCV)





- Select a commercially available TLCV resistant tomato variety. For instance, varieties from Southern India such as 'Sankranthi,' 'Nandi,' and 'Vybhav' have been reported to be resistant (Muniyappa et al. 2002). Consult the local extension agency for the availability of resistant or moderately resistant varieties.
- Destroy healthy and diseased tomato plants right after completing the final harvest and burn the crop residues.
- Whitefly is a polyphagous insect; it has several host plants for feeding and survival ranging from cultivated crops to weeds. The selected field for tomato planting or seedling production should be clean and located away from these host plants or weeds. If seedlings happen to be grown in those areas, cover the seedling beds or trays with insect-proof (50-64 mesh) nylon net.
- Use yellow sticky traps in the seedling production areas at the rate of 1-2 traps/50-100 m² to trap the whiteflies. Similarly, use yellow sticky traps or sheets (at least 10 per ha) in the tomato field.
- Neem formulations and imidacloprid (if recommended in your region) can be applied as soil drenching or foliar application to control whitefly in tomato seedling production.
- Plant tall border crops like maize, sorghum, or pearl millet to reduce whitefly infestations. Reflective plastic or straw mulches will reduce whitefly landings on tomato crops.

- Spray only those systemic pesticides that have been recommended by the local extension service to control whitefly in healthy and/or diseased tomato plantings to prevent further migration and subsequent spread of the disease. Do not use the same compound or pesticide group continuously to avoid the development of pesticide resistance in insects.





Tomato leaf miner

Liriomyza bryoniae Kaltenbach

(Diptera: Agromyzidae)

L. bryoniae is known to occur in China, India, Japan, Korea, Taiwan, and Vietnam in Asia, and Egypt and Morocco in Africa. However, *L. sativae* Blanchard, *L. trifolii* Burgess and *L. huidobrensis* Blanchard may also cause damage in tomato. *L. huidobrensis* occurs mostly in high elevations, whereas *L. sativae* occurs in low elevations (Spencer 1989; Shepard et al. 1998; Sivapragasam and Syed 1999; Rauf et al. 2000; Andersen et al. 2002; Andersen and Tran 2006). The host ranges of *L. huidobrensis* and *L. trifolii* each encompass more than 400 species of plants in 12 families (Reitz and Trumble 2002). *L. bryoniae* is a polyphagous insect and it has been recorded as damaging pest on several plant species in at least 16 families (Spencer 1990), although it prefers to feed on Cucurbitaceae. The major host plants are tomato, melon, watermelon, cucumber, cabbage, and lettuce.

Biology

L. bryoniae adult is a small, gray-colored fly. The adult males are slightly smaller in size than the females. Males appear to emerge prior to females (Parrella 1987). Females are more active than the males. The mesonotum is shiny black in color. The third leg segment (femur) is yellow in color, but the subsequent leg segments are brownish. Abdomen with tergites is yellow laterally. The longevity of adult female is 3-12 days (Cheng 1994). Males live for a shorter time than females (Parrella 1987). Each female can lay as many as 184 eggs in its lifetime (Lee et al. 1990).

Eggs are small, oval-shaped, white, and translucent. Eggs often resemble thrips eggs. Eggs are deposited through the upper or lower leaf surfaces. Eggs increase in size after oviposition, possibly through the imbibition of fluids from plant tissues (Parrella 1987). Egg period is about 3-7 days.

Larva is a cylindrical maggot, with tapering anterior end and truncated posterior end. The larva has four instars feeding inside the leaf tissues. The fourth instar occurs between puparium formation and pupation, which is rarely reported (Parrella 1987). The larval period is about one to two weeks.

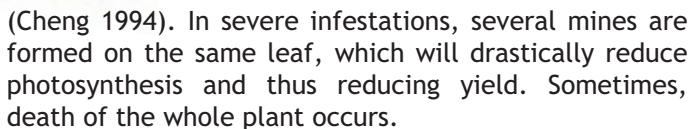
Most of the larvae move to soil for pupation from leaf mines by cutting a semi-circular slit, usually at or near the end of the mine. Pupa is also oval-shaped, yellow to brown in color. Pupal period is about 8-11 days (Parrella 1987).

Damage symptoms

The adult females damage mesophyll cells in a specific manner, creating either a tubular leaf puncture or a fan-shaped leaf puncture. Eggs are deposited in tubular leaf punctures. After every leaf puncture, the females back over the wounds and feed from them. Hence, the leaf punctures also can be considered feeding punctures. Males are unable to create their own punctures, and feed from punctures created by females. Leaf puncturing can reduce photosynthesis and may kill young plants (Parrella 1987).

The larva feeds the leaf mesophyll and causes irregular mines on leaf surfaces (Plate 16). The size of the mines and rate of mine formation increases with larval development (Parrella 1987). For instance, the leaf area mined by the last larval instar is about 88% of total larval infestation





- Yellow is the most attractive color to *Liriomyza* adults (Parrella 1987). Yellow sticky traps or cards may reduce the density of leaf miners. Traps also can be used to monitor populations in the field.
- Leaf miners have several parasitoids. For instance, *Gronotoma micromorpha* Perkins (larval-pupa parasitoid), *Chrysocharis pentheus* Walker, *Neochrysocharis formosa* (Westwood) and *Diglyphus isaea* Walker (larval parasitoids) and *Halticoptera circulus* Walker and *Opius phaseoli* Fischer (pupal parasitoids) are known to occur in Asia, including Japan, Malaysia, Sri Lanka, and Taiwan (Lee et al. 1990; Sivapragasam and Syed 1999; Niranjana et al. 2005; Abe 2006). Natural enemies help to keep the leaf miner population in check. Broad-spectrum chemical pesticides should be avoided, as these pesticides harm the parasitoid complex.
- Yield loss in tomato due to *L. bryoniae* infestation is not dependent on the severity of damage alone; but it is mainly dependent on the stage of the crop. For instance, a severe infestation (30 mines/leaf) on leaves adjacent to flowers at the time of half-swollen fruit size, results in 10% yield loss (Ledieu and Helyer 1985). The tomato crop in its early vegetative stage may not require any pesticide application to control *L. bryoniae* infestation.
- Leaf-miners rapidly can develop resistance against chemical pesticides. Resistance has been reported in *L. trifolii* (Parrella 1987), *L. huidobrensis* (Milla and Reitz 2005), and *L. sativae* (Hofsvang et al. 2005). Follow proper rotation and window periods.



Plate 16: Tomato plant infested by leaf miner showing the mines





Spider mites

***Tetranychus urticae* Koch, *T. cinnabarinus* Boisduval, *T. evansi* Baker & Pritchard**
(*Acarina: Tetranychidae*)

Spider mites emerged as serious pests of vegetable crops in South Asia, Southeast Asia, Africa, Europe and Mediterranean countries. Low relative humidity favors the multiplication of mites and precipitation is the only important abiotic factor that restricts spider mite populations.

Biology

T. urticae is commonly known as red spider mite or two-spotted spider mite. They are minute in size and vary in color, from green to greenish yellow, to brown or orange red with two dark spots on the body. Eggs are round, white- or cream-colored and egg period is 2-4 days. Upon hatching, it will pass through a larval stage and two nymphal stages (protonymph and deutonymph) before becoming adult. The lifecycle is completed in one to two weeks. There are several overlapping generations in a year. The adult will live up to three or four weeks.

T. cinnabarinus is commonly known as carmine spider mite. This is similar to two-spotted spider mite, but carmine in color.

T. evansi is also known as red spider mite, and it is similar to two-spotted spider mite. This is the predominant species in several countries in Africa on tomato and other solanaceous vegetables. In Asia, it has been reported in Taiwan.

Damage symptoms

Spider mites usually extract the cell contents from the leaves using their long, needle-like mouth part. This results in reduced chlorophyll content in the leaves, leading to the formation of several white or yellow speckles on the leaves (Plate 17). In severe infestations, leaves will completely desiccate and drop-off. The mites also produce webbing on the leaf surfaces (Plate 18); under severe conditions, the whole plant is wrapped in the webs (Plate 19). Under high population densities, the mites move to the tip of the leaf or top of the plant and congregate to form a ball-like mass (Plate 20), which will be carried to newer leaves or plants by wind.



Plate 17: White and yellow speckles caused by spider mites

Plate 18: Webbing of leaves by spider mites



Management

- Several predators of spider mites occur in most of the countries. For instance, *Stethorus spp.*, *Oligota spp.*, *Anthrocnodax occidentalis*, *Feltiella minuta*, etc. are known to occur in Taiwan (Ho 2000). Avoid using any broad-spectrum pesticides, as these may kill predators, thus leading to the outbreak of spider mites.
- Predatory mites such as *Phytoseiulus persimilis* and several species of *Amblyseius*, especially *A. womersleyi* and *A. fallacies*, can be used to control spider mites. They are more effective under protective structures and in high humidity conditions.
- Green lacewing (*Mallada basalis* and *Chrysoperla carnea*) are also effective generalist predators of spider mites. A third-instar grub of *C. carnea* can consume 25-30 spider mite adults per day; however, it needs supplemental food for long-term survival (Hazarika et al. 2001).
- Spray acaricides following local recommendations. Usually, the mectin group of pesticides (e.g., avermectin and milbemectin) are effective acaricides. However, continuous use may aggravate resistance in mites. Proper rotation and window periods should be followed.



Plate 19: Webbing of tomato plant by spider mites



Plate 20: Congregation of spider mites in the leaf tips



- Plant tall border crops like maize, sorghum, or pearl millet to reduce the infestation of whitefly.
- Destroy healthy or diseased tomato plants right after completing the final harvest and burn the crop residues.



Plate 21: Trap cropping with marigold to manage *H. armigera*



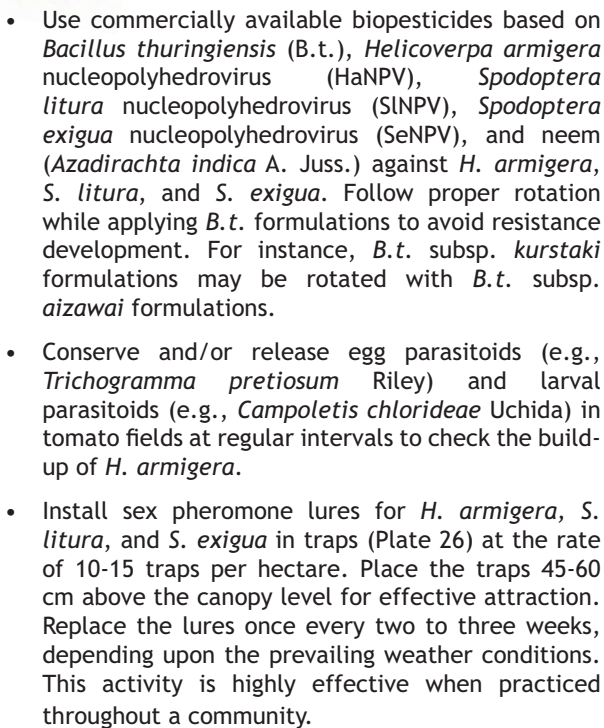


Plate 22: Trap cropping with castor to manage *S. litura*

Plate 23: Tomato seedling production under net tunnel

Plate 24: Production of tomato under net-house





- Do not spray broad-spectrum pesticides against early season sucking pests. This may disrupt the complex of natural enemies in the ecosystem and lead to a resurgence of sucking pests. If necessary, use systemic pesticides recommended by the local extension service in the region. Do not use the same compound or pesticide group(s) continuously to reduce the development of pesticide resistance in insects.

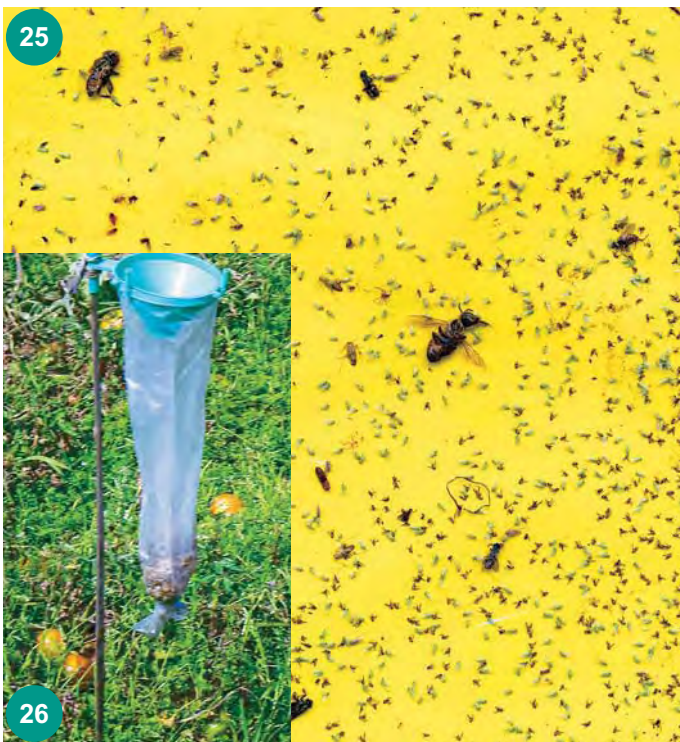


Plate 25: Yellow sticky card with trapped leaf miners (small-sized black-colored insects)

Plate 26: Sex pheromone traps to monitor and/or mass-trap *H. armigera* moths





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Managing bacterial diseases of tomato

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Bacteriology

AVRDC - The World Vegetable Center





Tomato bacterial spot

Pathogen

Bacterial spot can be caused by several species of *Xanthomonas*. The most common one is *X. euvesicatoria* (Jones et al. 2004), previously known as *X. axonopodis* pv. *vesicatoria* or *X. campestris* pv. *vesicatoria*, a motile, strictly aerobic, and gram negative rod with a single polar flagellum. On nutrient agar, single colonies that are circular, wet, shiny, and yellow can be observed two days after incubation at 30°C (Plate 27). There are several races reported. Some races can cause disease on both pepper and tomato, whereas some are specific to tomato or pepper.

Symptoms

The pathogen can affect all above-ground parts of tomato including stem, leaf, fruit, etc. (Plate 28). Lesions on leaves are small (< 3 mm in diameter), brown and circular with water-soaked margins. Lesions on tomato leaves can be confused with early blight, gray leaf spot, and target spot. Unlike early blight caused by *Alternaria solani*, bacterial spot lesions do not have concentric rings. The lesions are generally darker in color and less uniformly distributed than gray leaf spot. Under optimal conditions, lesions can coalesce to form dark streaks or blighting. Defoliation caused by bacterial spot is rare on tomato. Fruit lesions begin as raised blisters. Older lesions are brown, scab-like and sunken in the middle.

Disease cycle and epidemiology

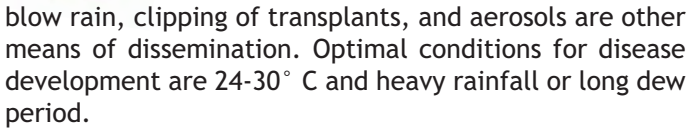
The pathogen is able to survive on volunteer plants and diseased plant debris. Seed may serve as a medium for the survival and dissemination of the bacterium. Wind-



Plate 27: Colonies of *X. euvesicatoria* culture on 523 (left) and NA (right) medium after incubation at 30°C for 2 days



Plate 28: Lesions on tomato seedlings (a), stem (b), leaves (c), and fruits (d)



- Rotation to avoid carryover of inoculum on volunteer plants and plant debris.
- Use pathogen-free seeds and transplants.
- Use seed treatment, such as soaking in diluted Chlorox (1% sodium hypochlorite) for 5 minutes.
- Use copper or copper+maneb sprays.
- Resistant cultivars are available for tomato, but they may not be resistant to all strains.

Ralstonia solanacearum (= *Burkholderia solanacearum* = *Pseudomonas solanacearum*) is gram negative rod, motile with one or more polar flagella, and aerobic. On plates of tetrazolium chloride medium (TTC), a differential medium for *R. solanacearum*, large fluidal colonies with pink or red centers are distinguishable and most of them are virulent (Plate 29b). The pathogen is a species complex with wide variation in host range (used for race grouping), utilization of carbohydrates (used for biovar grouping), aggressiveness, genetic fingerprints, etc.

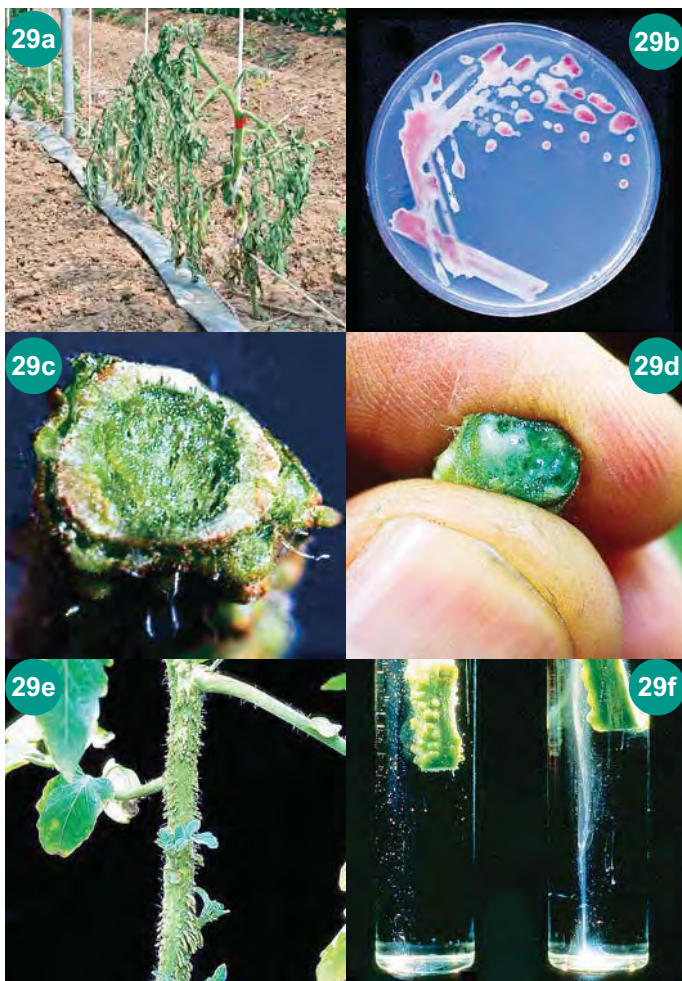


Plate 29: Infected plant displays: (a) wilting but not yellowing of leaves; (b) colonies of *R. solanacearum* on TTC medium after incubation at 30°C for 48 hours; (c) vascular browning; (d) whitish bacterial mass, observed when pressing the cut site; (e) adventitious roots and leaf epinasty; (f) bacteria streams from an infected stem, but not from a healthy stem.



Symptoms

Typical symptoms of the disease first appear as drooping of a few young leaves. A sudden complete wilt soon follows. Most of the time, leaves are still green when the plants wilt (Plate 29a). Adventitious roots and leaf epinasty are more obvious on tomato when the disease develops slowly under unfavorable climatic conditions or on resistant varieties (Plate 29e). The vascular system of infected stems appears brown and becomes darker as the disease progresses (Plate 29c). Wilting in tomato can be caused by fungal pathogens, root-knot nematode, or excess soil moisture. To distinguish bacterial wilt from other vascular diseases, a clean stem section from wilted plants can be suspended in water; a milky stream of bacterial cells would flow from xylem elements, 3 to 5 minutes later (Plate 29d and f).

Disease cycle and epidemiology

R. solanacearum has a wide host range (> 200 cultivated species and weeds). The pathogen can colonize many weed species without showing symptoms. This may enhance the long-term survival of the pathogen in the soil. *R. solanacearum* can survive in the soil for a long period depending on the environment. Poorly-drained soils, moderate to high soil temperature, and a low to moderate pH are conducive to its survival.

The bacterium enters the host through natural wounds in the roots, wounds made during transplanting, or those caused by insects, nematodes etc. Infection and disease development are favored by high temperature (30-35°C) and high soil moisture. The bacterium returns to soil with decaying diseased plant parts and is disseminated by free water, soil movement, or the movement of infected transplants.

Control measures

- Use pathogen-free seedbeds to produce clean transplants. This can be assisted by fumigating seedbeds and pasteurizing the potting mixture.
- Rotate with non-host crops, particularly with paddy.
- Resistant varieties for tomato are available, but their reaction may be different from location to location.
- Use resistant tomato or eggplant varieties as rootstocks to increase the resistance level of scion varieties (Plate 30).



Plate 30: Integrating resistant eggplant rootstock and the soil amendments of urea and lime mixture (a) or Indian mustard tissues (b) to control bacterial wilt in tomato production fields. Non-grafted tomato plants showed wilting symptoms.





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Managing fungal diseases of tomato

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Mycology

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Damping-off

Caused by:

- (i) *Pythium aphanidermatum*, *P. ultimum*, *Phytophthora capsici*, *P. parasitica* (Phylum: *Heterokontophyta*, Class: *Oomycetes*, Order: *Peronosporales*, Family: *Pythiaceae*)
- (ii) *Rhizoctonia solani* (Phylum: *Basidiomycota*, Class: *Basidiomycetes*, Order: *Polyporales*, Family: *Corticaceae*)

Symptoms

The pathogen may cause seed rot, pre-emergence damping-off, post-emergence damping-off (Plate 31) and stem rot (Plate 32). The decay of seedling before it emerges through the soil is called pre-emergence damping-off. The post-emergence phase of the disease either begins as a dark-colored and soft lesion around a major portion of stem or causes a brown, reddish brown or nearly black lesion near the soil line. The infected seedling falls over, withers and dies. It usually occurs in patches in nursery beds or in scattered areas of the field in direct-seeded crops. In heavily affected soils, a general death of seedlings may be observed.

Conditions for development

Seedlings are particularly susceptible during first two weeks after sowing. Overwatering, overcrowding, or poor ventilation of seedling nursery favor disease development, as do cool, damp, cloudy weather or poorly drained areas of field, cool and warm temperatures, poor-quality light, or unbalanced nutrition, particularly excessive nitrogen.

Management

- High quality seed and a chemical- or heat-pasteurized planting medium should be used.
- Excessive watering and poorly drained areas of field should be avoided.
- Use high beds: more than 35 cm of height is better for water drainage. If possible, plant when the soil is warm as seed germinates faster, and seedlings become more vigorous and thus less susceptible to the disease.
- Apply 35% Etridiazole WP 3,000X just after transplanting the seedlings by drenching to reduce the pathogen population in each transplanting hole. In field plantings a fungicide seed treatment may help prevent serious damping-off problems.



Plate 31: Infected by *P. aphanidermatum* and caused seedling post-emergence damping-off.

Plate 32: Infected by *Pythium* spp. and caused stem rot after transplanting.





Early blight

Caused by: *Alternaria solani* (Phylum: *Ascomycota*, Class: *Dothideomycetes*, Order: *Pleosporales*, Family: *Pleosporaceae*)

Symptoms

Early blight occurs on the foliage, stem, and fruit of the tomato plant and can cause severe damage during all stages of development of the plant. It is often observed in the field as small brownish-black lesions on the older foliage. The tissue surrounding a spot may become yellow, and when spotting is abundant the entire leaf may become yellow. The spots enlarge rapidly; concentric rings may be distinguished in the dark brown portion of the spot (Plate 33). Stem lesions are small, dark, and slightly sunken in the beginning. They enlarge to form circular or enlarged lesions with pronounced concentric rings and light centers (Plate 34). The fruit becomes infected, generally through the calyx or stem attachment, in either the green or ripe stage. The fruit lesions attain considerable size, occasionally involving the entire fruit, and usually have concentric ringing (Plate 35). Uncontrolled, the disease may cause severe defoliation, resulting in reduced fruit number and size and marketable fruits.

Conditions for development

The pathogen can be seed-borne; it also persists in the soils on crop debris or on volunteer tomatoes and wild solanaceous plants. The fungus is spread rapidly by wind, rain, and farm equipment. Extended periods of leaf wetness from dew or rain favor disease development; frequent rain or overhead irrigation may also promote disease development. Stressed plants are most

susceptible, *e.g.*, when they are attacked by nematodes and also during fruiting.

Management

- Use resistant or tolerant cultivars.
- Use treated seeds and pathogen-free seedlings.
- Pasteurize the soil in seedbeds (with steam, fumigants, or solarization), or use pathogen-free artificial soil mixes.
- Use long crop rotations, eradicate weeds and volunteer tomato plants, fertilize properly, and keep the plants growing vigorously.
- Avoid planting overlapping crops in adjacent areas.
- Spray regularly with fungicides, *e.g.*, copper hydroxide if necessary.



Plate 33: Concentric rings appear in the dark brown lesion.

Plate 34: Enlarged lesion with concentric rings and light center appear on stem.

Plate 35: Concentric rings appear on green or ripe fruit.



Late blight

Caused by: *Phytophthora infestans* (Phylum: *Heterokontophyta*, Class: *Oomycetes*, Order: *Peronosporales*, Family: *Pythiaceae*)

Symptoms

The fungus attacks all above-ground parts of the plant:

Leaf lesions: Irregular water-soaked patches (Plate 36) may expand to encompass large areas of the leaf. White fungal sporulation may be observed on underneath side; later the lesions dry and turn brown. Blighting of the entire foliage may occur.

Stem lesions: At first, irregular water-soaked areas may progress and kill sections of stems and petioles (Plate 37), or they may remain superficial and dry out to form dark brown lesions.

Fruit lesions: Firm, olive to brown irregular shaped areas that cause fruit to have a rough leathery surface (Plate 38); lesions may enlarge to encompass the entire fruit.

Conditions for development

Sporangia develop on the leaves, spreading through the crop during cold to moderate temperatures (10-25°C) and humidity greater than 75% for two days or more, especially during extended periods of leaf wetness from frequent rain or dew formation. Free water on plant surfaces is required for the germination and penetration of spores in the plant tissues. The fungus persists on tomato and potato plants and in potato tubers; it does not survive saprophytically. The new genotype attacks both tomato and potato. Sporangia are produced on infected tissues and are dispersed by wind and splashing rain.

Management

- Use resistant cultivars such as ‘Hualien ASVEG 17.’
- Use disease-free seedlings.
- Avoid planting tomatoes near potatoes.
- Apply effective fungicides, e.g., Famoxadone + Cymoxanil, Azoxystrobin, and Dimethomorph, etc.
- Eliminate all potato cull piles in the vicinity of tomato plantings and destroy volunteer potato plants that grow from overwintered tubers.
- Rain shelters can reduce disease severity.



Plate 36: Irregular water-soaked patches appear on leaves.

Plate 37: Irregular water-soaked areas that progressed and killed the petioles and stems.

Plate 38: Firm, olive to brown irregular shaped areas that caused fruit to have a rough leathery surface.



without a host. Three races are known to exist in the world and race 2 is predominant in Taiwan.



Plate 39: Significant yellowing of the lower leaves appear only one side of the plant.

Plate 40: Plants begin to wilt in the top and wilting becomes progressively worse until plants are permanently wilted.

Plate 41: The browning of the vascular system.

Management

Southern blight

Caused by: *Sclerotium rolfsii* (imperfect stage); *Athelia rolfsii* (perfect stage) (Phylum: *Basidiomycota*, Class: *Basidiomycetes*, Order: *Atheliales*, Family: *Atheliaceae*)

Symptoms

Symptoms usually appear on plant parts in, on, or near the soil. Plants of any age may be attacked. The most common symptom is a brown to black rot of the stem, which develops near the soil line (Plate 42). The lesion develops rapidly, completely girdling the stem and resulting in a sudden and permanent wilt of all above-ground parts. Young plants may fall over at the soil line. Under moist conditions, an abundance of white, robust mycelium develops on the lesion and sometimes extends several centimeters up the stem of a mature plant. Tan to reddish brown, spherical sclerotia appear on the mycelial mat (Plate 43) after few days. The fungus quickly penetrates the epidermis of fruit in contact with infested soil. White mycelium and developing sclerotia quickly fill the lesion cavity.

Conditions for development

High temperatures and high soil moisture favor disease development; wilting may be more evident when soils begin to dry out. Dense planting and frequent irrigation promote infection. Sclerotia serve as the principle overwintering structures and primary inoculum for disease. Sclerotia can survive in soil for several years on crop debris. The fungus has an extremely broad host range. It is spread by surface water or movement of infested soil.





Management

- The most effective control for this disease is a good sanitation program. Removal and burning of all infected plants prevents disease build-up. Deep plowing also serves as a cultural control tactic by burying sclerotia and infected plant debris deep in the soil.
- A crop rotation program of at least three years with non-susceptible crops such as corn, small grains, or sorghum can reduce losses.
- Fumigation of high value crops is also effective in controlling southern blight. Some soil fungicides offer some protection, e.g., Etridiazole.
- Use solarization with polyethylene sheeting.
- Use biological control with certain antagonistic fungi (e.g. *Bacillus subtilis*, *Gliocladium virens*, *Trichoderma harzianum*, *Trichoderma viride*).
- Use plastic mulch or stake the plant to reduce soil splash.
- Add soil amendments in acid soil before planting.
- Use resistant cultivars when available.





Plate 42: An abundance of white, robust mycelium develops on the stem near the soil line.



Plate 43: Tan to reddish brown, spherical sclerotia appear on the white mycelial mat.





Black leaf mold (*Cercospora* leaf mold)

Caused by: *Pseudocercospora fuligena* (Phylum: Ascomycota, Class: Dothideomycetes, Order Capnodiales, Family: Mycosphaerellaceae)

Symptoms

Yellow spots appear on the upper leaf surface at early stage of infection; gray to black conidial sporulation occurs predominantly on the lower leaf surface (Plate 44) under humid conditions. Numerous lesions may coalesce to cover most of leaf surface. Leaves roll upward, exposing the lower surface and begin to dry (Plate 45); many of these leaves remain attached on the plants and have a soot-covered appearance. Most of the plant foliage can be killed, but fruit is not infected.

Conditions for development

The pathogen over-seasons on old diseased plants or other hosts, e.g., black nightshade (*Solanum nigrum* L.). Disease development is favored by high humidity, long periods of leaf wetness, and by moderate to high temperatures (27°C). Leaf wetness from dew formation is most effective for disease development; heavy rainfall may reduce disease severity. Conidia are the major inoculum and can be disseminated by air currents, splashing rain, running water, or machinery.



Plate 44: Gray to black conidial sporulation occurs predominantly on the lower leaf surface.

Plate 45: Numerous lesions may coalesce to cover most of the leaf surface. Leaves roll upward exposing the lower surface and begin to dry.



Management

GLOSSARY

Acaricides	Pesticides that kill mites. They are also known as miticides
Adsorbed	The process by which atoms, molecules, or ions are taken up from the soil solution or soil atmosphere and retained on the surfaces of soil solids or organic matters by chemical or physical binding. Adsorption occurs in many natural physical, biological, and chemical systems.
Ammonia volatilization	Ammonia volatilization refers to the loss of NH_3 from the soil as a gas to the atmosphere and is normally associated with high free NH_3 concentrations in the soil solution and high soil pH. Surface applications of ammoniacal fertilizers or readily decomposable organic wastes to soils can result in considerable N loss by NH_3 gas volatilization, particularly if the soil is alkaline in nature.
Algal blooms	A sudden, massive increase or accumulation in the population of algae in lakes and streams, often resulting in high turbidity and green- or red-colored water, and commonly stimulated by nutrient enrichment with phosphorus and nitrogen. Algal blooms can adversely affect the health of people and marine ecosystems and the viability of local and regional economies.

Antioxidant	A substance or molecule that prevents or slows down the oxidation of other molecules. Oxidation produces by-products (free radicals) that can cause cell damage.
Atrophied	Diminution of a body part, organ or tissue.
Balanced fertilizer application	Application of fertilizers using a balanced combination of organic and inorganic sources, and a balanced ratio among N, P and K nutrients. It sustains good soil fertility conditions and makes the nutrient supply more efficient.
Basal application	Applying organic and/or inorganic fertilizers before sowing or transplanting of plants. Basal applications can be broadcast, applied in bands, or placed close to the plants.
Broad-spectrum pesticide	Non-selective pesticide that could kill a wide range of species in an ecosystem.
Canopy	Uppermost foliage cover in a plant or tree.
Cation exchange capacity (CEC)	The capacity of soil to hold nutrients for plant use. Specifically, CEC is the amount of negative charges available on clay and humus to hold positively charged cations. Effective cation exchange capacity (ECEC) is reported for acid soils (pH<5). It is usually expressed as centimoles of charge per kilogram of soil (cmol _c /kg).

Chlorophyll	Green pigment that gives green color to a plant's leaves, stems, etc. It is vital for photosynthesis.
Chlorotic	Chlorotic (adjective); chlorosis (noun) relates a condition in plants showing discoloration of normally green plant parts caused by disease, lack of nutrients, or various air pollutants. Chlorotic leaves range from light green through yellow to almost white.
Cocoon	A pupal case, usually made up of silk.
Compost	Organic residues, or a mixture of organic residues and soil, that have been mixed, piled, and moistened, with or without addition of inorganic fertilizer, and generally allowed to undergo biological decomposition until the original organic materials have been substantially altered or decomposed. Composting converts organic constituents, usually wastes, into humus-like material suitable for use as a soil amendment or organic fertilizer.
Denitrification	Biochemical reduction of nitrate or nitrite to gaseous nitrogen, either as molecular nitrogen or as an oxide of nitrogen, by bacterial activity or by chemical reactions involving nitrite.

Desertification	The process of the gradual transformation of habitable land into desert. Land in arid, semi-arid, and subhumid areas deteriorates due to loss of vegetation and soil moisture. It is usually caused by human mismanagement of land resources, with drought and other natural events exacerbating the effect of human action.
Dessicate	Become dried.
Deutonymph	The second nymphal stage in mites. Mostly, the deutonymph molts into adult stage.
Diapause	Temporary suspension of development (physiological dormancy) due to adverse environmental conditions.
Dorsal	Located at the back or upper surface of an insect body.
Eutrophication	A condition in an aquatic ecosystem where high nutrient concentrations cause algae to bloom and grow rapidly which leads to a deficiency of oxygen in the water. Algae blooms disrupt the ecosystem in two ways. First, algae blocks sunlight that is needed by the underwater grasses which feed fish and other animals. Second, when the algae die and decompose, they use up oxygen in the water that is needed by fish and other animals. Human activities such as agriculture runoff, urban runoff, and sewage leakages accelerate these increased nutrient levels.

Exuvia (also exuvium; plural, exuviae)	Cast skin or shed skin after moulting.
Flagellum	Hair-like, whip-like, or tinsel-like appendage of a motile cell, bacterium or zoospore that provides locomotion.
Honey dew	Liquid feces of insects in the order Homoptera, containing soluble sugars and amino acids.
Inoculum	Pathogen or its parts, capable of causing infection when transferred to a favorable location.
Inorganic fertilizers	Fertilizers composed of synthetic chemicals and/or minerals. Chemical or inorganic fertilizers that are not obtained from a source that is or has been alive.
Instar	Developmental stage during the larval period. The time interval between two subsequent moulting is known as stadium. The form of an insect during any stadium is called as instar.
Leaf mines	Feeding tunnels between epidermal layers of the leaf.
Mesonotum	Upper surface of the mesothorax (middle of the three segments in the thorax of an insect).

Nitrogenous	Chemical compounds (usually organic) containing nitrogen in combined forms. Proteins and nitrates are nitrogenous compounds.
Nymph	Larval stage in certain insect orders including Hemiptera. Nymphs will directly develop into adults without undergoing pupal stage.
Organic fertilizers	Naturally-occurring organic materials that contain sufficient plant nutrients to be of value as fertilizers, (e.g. manure, green manure, worm castings, slurry, peat, seaweed, etc.), or naturally occurring mineral deposits (e.g. saltpeter, guano). Processed organic fertilizers include compost, bone meal, humic acid, amino acid, brassin and seaweed extracts. Decomposing crop residue from prior years is another source of organic fertilizer. In addition to increasing yield and fertilizing plants directly, organic fertilizers can improve the biodiversity of soil organisms and increase soil organic matter to sustain long-term productivity.
Parasitoid	An organism that spends most of its lifecycle within a single host insect for nutrients and protection and ultimately kills it.
Pathogen	An entity, usually a microorganism, that can incite disease.

Sclerotium (plural: sclerotia)	A compact mass of hyphae with or without host tissue, usually with a darkened rind, and capable of surviving under unfavorable environmental conditions.
Skeletonization	Process of destroying the tissues and leaving the basic structure or shape (of the leaves).
Soil buffering capacity	The ability of solid phase soil materials to resist changes in ion concentration in the solution phase. It includes pH buffering capacity as well as the buffering capacity of other ionic and molecular components. pH buffering capacity refers to the ability of a soil to resist large changes in pH when an acidic or alkaline solution is added to it.
Solanaceous	Plant species in the family Solanaceae.
Sooty mold	Black powdery coat on the leaf surfaces due to the growth of saprophytic fungi on the excretions (honey dew) of certain homopteran insects.
Sporangium (plural: sporangia)	A container or case of asexual spores. In some cases it functions as a single spore.
Sporulation	The process of spore production.

Starter Solution	Inorganic fertilizer solution with very high NPK concentration (>200-240 mg N/plant) applied near plant roots immediately after transplanting. The volumes applied each time are small (about 50 ml) so the solution can be held near the roots. Starter solution can tentatively build up high nutrient gradients in the soil solution around the soil-rhizosphere system. It provides young transplants with readily available nutrients before their root systems are well-established, enhances initial growth of vegetables, and improves fertilizer use efficiency.
Systemic pesticide	Pesticide that move throughout the plant system after application.
Tergites	Hardened plates on the upper surface of thorax and abdomen of an insect.
Thoracic	Located in the thorax (chest).
Venation	The arrangement of veins in a leaf.
Virulent	Highly pathogenic; having the capacity to cause severe disease.

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